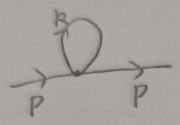
"耦合常数量纲为负则理论不可重整" = 此公说此是一一

第十章 刘心怡

题1. 计算ϕ^4的重整化.

1) 计算单圈自能图修正，得到m_0与m的关系，算Z. 抵消项.



$$-i\Sigma_1(p^2) = \frac{1}{2}(-i\lambda_0)\int\frac{d^Dk}{(2\pi)^D}\frac{i}{k^2-m_0^2+i\epsilon}$$

$$\Sigma_1(p^2) = \frac{\lambda_0}{2}\int\frac{d^Dk}{(2\pi)^D}\frac{i}{k^2-m_0^2+i\epsilon} \qquad D=4-2\epsilon$$

附录给出 $I_1(1) = \int\frac{d^Dk}{i\pi^{D/2}}\frac{1}{k^2-m^2} = (-1)\Gamma(\epsilon-1)(m_0^2)^{1-\epsilon}$

$$\therefore \Sigma_1 = \frac{\lambda_0}{2}\cdot i\frac{1}{2^D\pi^{D/2}}\cdot i\, I_1(1) = -\frac{\lambda_0}{2^{D+1}\pi^{D/2}}I_1(1) = +\frac{\lambda_0}{2}\frac{1}{(4\pi)^{D/2}}\Gamma(\epsilon-1)(m_0^2)^{1-\epsilon}$$

$$= \frac{\lambda_0}{2}\frac{1}{(4\pi)^{D/2}}m_0^2\left[-\frac{1}{\epsilon}-1+\gamma_E+\log(m_0^2)\right]$$ (γ_E为欧拉-马歇罗尼常数)

在$\epsilon\to 0$时，$\Sigma_1 = \frac{\lambda_0 m_0^2}{32\pi^2}\left[-\frac{1}{\epsilon}-1+\gamma_E+\log(m_0^2)\right]$ ① 有$1/\epsilon$发散.

计算完整传播子：

$$\frac{i}{k^2-m_0^2}+\frac{i}{k^2-m_0^2}(-i\Sigma_1)\frac{i}{k^2-m_0^2}+\cdots$$

$$= \frac{i}{k^2-m_0^2}\left(1+\frac{\Sigma_1}{k^2-m_0^2}+\left(\frac{\Sigma_1}{k^2-m_0^2}\right)^2+\cdots\right) = \frac{i}{k^2-m_0^2}\frac{1}{1-\frac{\Sigma_1}{k^2-m_0^2}} = \frac{i}{k^2-m_0^2-\Sigma_1}$$

$\therefore m^2 = m_0^2+\Sigma_1$ ② m_0和Σ_1中的发散项抵消

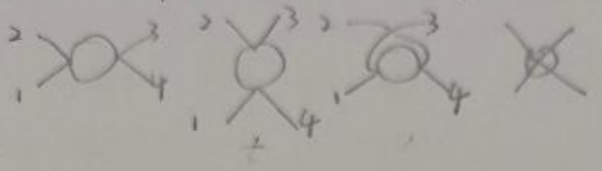
$$Z = \frac{1}{1-\frac{d\Sigma_1}{dp^2}}\Big|_{p^2=m^2} = 1 \quad ? \quad ③$$

那么$\delta Z = 0$. 抵消项.

$\mathcal{L}_{抵消项} = -\frac{1}{2}\delta m\phi^2$ $\qquad \delta m = m_0^2-m^2 = -\Sigma_1 = -①$

重出"正确吗? why?"

2) 计算四点振幅单圈修正. [8A]

其费曼图为  和树图 $= -i\lambda$

则

四点振幅 $-iM_4 = -i\lambda \underbrace{- i\Lambda^{(1)}(s,t,u;m,\lambda)}_{\text{单圈图的振幅}} - i\delta\lambda$ ①

则 s-channel 的贡献 $= (-i\lambda)^2 \cdot \frac{1}{2} \cdot \int d^4k/(2\pi)^4 \cdot \frac{i}{k^2-m^2+i\epsilon} \frac{i}{(k+p)^2-m^2+i\epsilon}$

$p \to p_1+p_2$

$= (-i)\cdot \frac{\lambda^2}{2} \cdot \underbrace{i\int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2-m^2+i\epsilon} \frac{1}{(k+p)^2-m^2+i\epsilon}}$ → 令其为 $I(s,m)$ ②

则同样的, $I(t,m) =$ 上式$(p \to p_1+p_3)$ $I(u,m) =$ 上式$(p \to p_1+p_4)$

即 $\Lambda^{(1)}(s,t,u;m,\lambda) = \frac{\lambda^2}{2}\cdot\left(I(s,m)+I(t,m)+I(u,m)\right)$ ③

用维度正规化. 令 $D = 4-2\epsilon$

则 $I(s,m) = i\int \frac{d^Dk}{(2\pi)^D} \frac{1}{k^2-m^2+i\epsilon} \frac{1}{(k+p)^2-m^2+i\epsilon}$

附录 $\frac{-1}{2\pi^{D/2}}\int \frac{d^Dk}{i\pi^{D/2}} \frac{1}{k^2-m^2} \frac{1}{(k-(-p))^2-m^2}$

$= \left[\frac{1}{\epsilon} - \gamma_E - \log(p^2) - \int_0^1 da_1 \log\left(\overset{-a_1^2+a_1}{-\frac{m^2}{p^2}}\right) + O(\epsilon)\right]\cdot\left(\frac{-1}{2\pi^{D/2}}\right)$ ④

$\underline{\epsilon \to 0}$ $-\frac{1}{16\pi^2}\cdot\left[\frac{1}{\epsilon} - \gamma_E - \log(p^2) - \int_0^1 da_1 \log\left(-a_1^2+a_1-\frac{m^2}{p^2}\right) + O(\epsilon)\right]$ ⑤

时 $\frac{1}{\epsilon}$ 发散.

Now, $I(p^2, m) = \frac{1}{16\pi^2} \cdot (\mathrm{Div} + \mathrm{Fin}(p^2, m, a_1))$ ⑥

Then. $-iM_4 = -i\lambda - \frac{i\lambda^2}{32\pi^2}(3\mathrm{Div} + \mathrm{Fin}(s) + \mathrm{Fin}(t) + \mathrm{Fin}(u)) - i\delta\lambda$ ⑦

引入重整化条件, satisfying that. $M_4 = \lambda$ (可以取特定的 (s_0, t_0, u_0) $\frac{4m^2}{3}$)

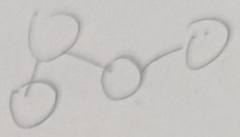
则 ⑦ $\Rightarrow -i\lambda = -i\lambda - \frac{i\lambda^2}{32\pi^2}[3\mathrm{Div} + \mathrm{Fin}(s_0) + \mathrm{Fin}(t_0) + \mathrm{Fin}(u_0)] - i\delta\lambda$

$$\delta\lambda = \frac{-\lambda^2}{32\pi^2} \cdot (3\mathrm{Div} + \mathrm{Fin}(s_0) + \mathrm{Fin}(t_0) + \mathrm{Fin}(u_0))$$

题2.(1) 证明D维 ϕ^n 理论的图满足 $L=P-V+1$ 圈 内线 顶点 $nV=E+2P$

$L=P-V+1$:

内线由圈被各顶点分割开来 (that means 1个圈有n个顶点就贡献n条内线) 以及连接圈与圈的线组成.



so $P-V$ = 连接圈之间的线数, 那么 $L=P-V+1$

$nV=E+2P$:

一个 External lines 贡献一个 vertice. 一个 internal propagators 贡献2个 vertices、ϕ^n 理论应该是1个 vertice 有n条线.

∴只考虑D而忽略中间的连接 $\Rightarrow nV=E+2P$.

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(2) 证: 对于QED, 每个圈贡献 $\int d^4p$; P_e 贡献 $\sim\frac{1}{p}$ P_γ 贡献 $\frac{1}{p^2}$

$\therefore pow(L)=4L-P_e-2P_\gamma=4(P_e+P_\gamma-V+1)-P_e-2P_\gamma$

$=3P_e+2P_\gamma-4V+4=3P_e+2P_\gamma-4(E_\gamma+2P_\gamma)+4$

$=4+3P_e-6P_\gamma-4E_\gamma \xlongequal[E_\gamma-\frac{1}{2}E_e]{P_e-2P_\gamma=} 4+3(E_\gamma-\frac{1}{2}E_e)-4E_\gamma$

$=4-E_\gamma-\frac{3}{2}E_e$ 证毕.

对于QCD: 根据Feynman rules, $L:\int d^4p$, $P_q:\frac{1}{p}$, $P_c:\frac{1}{p^2}$, $P_g:\frac{1}{p^2}$

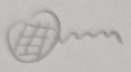
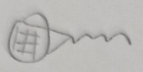
$V_{3g}:P$ $V_c:P$ so: $pow(L)=4L+V_{3g}+V_c-P_q-2P_c-2P_g$

利用公式 $L=(P_g+P_q+P_c)-(V_{3g}+V_c+V_q+V_{qg})+1$, $E_g+2P_g=3V_{3g}+4V_{4g}+V_q+V_c$

$E_q+2P_q=2V_q$ $4V_q=2E_q+4P_q$, $E_c+2P_c=2V_c$ $3V_c=\frac{3}{2}E_c+3P_c$

$3V_{3g}+4V_{4g}=E_g+2P_g-\frac{1}{2}E_q-P_q-\frac{1}{2}E_l-P_l$ 可证.

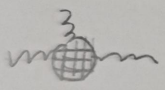
题3. (a) is badly divergent! However, 这个 vacuum bubbles 在计算 S 矩阵时会自动消失, 因此不予考虑它.

(b)  $\rightarrow$  $= ie\int d^4x\, e^{-iq\cdot x}\langle\Omega|T j_\mu(x)|\Omega\rangle$

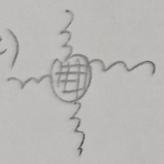
为保证 lorentz invariance, 上式=0, 不贡献

(c) $\mathrm{pow}(L)=4-2=2.$

but $it=(g^{\mu\nu}q^2-q^\mu q^\nu)\,\Pi(q^2)$ 如果存在 $C\cdot\Lambda^2 g_{\mu\nu}$ 二次发散, 则不变 ward 恒等式允许. 降阶圆积分最高阶的 log 发散.

(d)  $\mathrm{pow}(L)=4-3=1.$

根据 Furry 定理, 任何奇数个光子的闭合的费米子圈贡献精确为0

(e)  $D=0$ 对数发散.

(f) $\mathrm{pow}(L)=4-\frac{3}{2}\times 2=1$ However, $i\Sigma_2(\not p)=(-ie)^2\int\frac{d^4p}{(2\pi)^4}\gamma^\mu\frac{i(\not k+m)}{k^2-m^2+i\epsilon}\gamma_\mu\frac{-i}{(k-p)^2+i\epsilon}$

$$=-\frac{\alpha}{2\pi}\int_0^1 dx\left[(2-\epsilon)x\not p-(4-\epsilon)m\right]\left[\frac{2}{\epsilon}+\ln\frac{\tilde{\mu}^2}{(1-x)(m^2-p^2x^2)}\right]$$

log 发散.

(g) $\mathrm{pow}(L)=4-1-\frac{3}{2}\times 2=0$ log 发散

题4. 1) 证："耦合常数量纲为负的理论不可重整" = 耦合常数量纲和顶点数V前面系数为相反数. (2) "耦合常数无量纲的理论一定可重整"正确吗? why?"

(1) 证: $[g] = D - [O]$ → 相互作用项去掉g以外的算符 $= 4 - n_\phi[\phi] - n_\psi[\psi] - \delta[\partial_\mu] = 4 - n_\phi - \frac{3}{2}n_\psi - \delta$ ①

而 $pow(L) = DL - \sum_i p_i + \sum_j s_j$ ② $\quad p_i = \begin{cases} 2P & \text{for bosons} \\ P & \text{for fermions} \end{cases}$

而 $L = P - V + 1$ ③

简化一下吧!

则 ② $= 4(P-V+1) - 2P_B - P_F + V\delta = 4 + 2P_B + 3P_F - 4V + V\delta$ ④

而 $Vn_\phi = 2P_B + E_B$ ⑤ $\quad Vn_\psi = 2P_F + E_F$ ⑥

则 ④ $= 4 + (Vn_\phi - E_B) + \frac{3}{2}(Vn_\psi - E_F) - 4V + V\delta$

$= 4 - E_B - \frac{3}{2}E_F + V\underbrace{(n_\phi + \frac{3}{2}n_\psi + \delta - 4)}_{-①}$ ⑦

则 $pow(L) = 4 - E_B - \frac{3}{2}E_F - V[g]$

(i) $[g] = 0$ $\quad pow(L)$ 与V无关、可重整.

(ii) $[g] > 0$ $\quad pow(L) = 4 - E_B - \frac{3}{2}E_F - V[g]$ $\quad V\uparrow \to pow(L)\downarrow$, 即高阶费曼图会急收敛, 可重整.

(iii) $[g] < 0$ $\quad pow(L) = 4 - E_B - \frac{3}{2}E_F + V|[g]|$ $\quad V\uparrow \to pow(L)\uparrow$ 不可重整

证毕.

(2) 不正确. 它只考虑了表观发散, 而应还有很多其它原因. 如上一题一样. 例如是具有坏范的常规场论 (自拟)

重整化群作业

刘心怡

题1. 已知g满足 $u\frac{dg}{du}=\beta(g)$，其中β函数的微扰展开为

$\beta(g)=b_2g^2+b_3g^3+O(g^4)$. 考虑 $g'=g+cg^2+O(g^3)$. b_2. b_3. c是已知量.

计算g'的对应的$\beta(g')$的微扰展开表达式：$u\frac{dg'}{du}=\beta'(g')$. $\beta'(g')=?$

要求展开到g'前两阶. 并和$\beta(g)$比较展开前两阶的系数.

$$\frac{dg'}{du}=\frac{dg}{du}+2cg\frac{dg}{du}+\frac{dg}{du}O(g^2)=\frac{dg}{du}\left(1+2cg+O(g^2)\right)$$

$$\therefore \beta'(g')=u\frac{dg}{du}\left(1+2cg+O(g^2)\right)=\beta(g)\left(1+2cg+O(g^2)\right)$$

$$=\left(b_2g^2+b_3g^3+O(g^4)\right)\left(1+2cg+O(g^2)\right)$$

$$=b_2g^2+(2cb_2+b_3)g^3+O(g^4) \quad ①$$

再把①按g'代入，可求g与g'的关系. 由于 $g'=g+cg^2+O(g^3)$

令 $g=g'+a_2(g')^2+a_3(g')^3+O(g'^4)$

代回. $g'=(g'+a_2(g')^2+a_3(g')^3)+c(g'+a_2(g')^2)^2+\underbrace{O(g^3)}_{O(g')^3}+O(g'^4)$

$$=g'+(a_2+c)(g')^2+\underline{(a_3+2ca_2)(g')^3}+\underbrace{O(g')^3}_{\text{含有}kg'^3}+O(g'^4)$$

$\Rightarrow a_2=-c$ $\qquad a_3=2c^2-k$.

则 $g=g'-c(g')^2+(2c^2-k)(g')^3+O(g'^4)$ ②

将②代入①，$\Rightarrow b_2(g'^2-2c(g')^3)+(2cb_2+b_3)(g')^3$

$$\beta(g')=b_2g'^2+b_3(g')^3+O(g'^4) \Rightarrow \boxed{\text{系数不变}}$$

题2. 解. 按含常数展开 $r_0 = \sum_i \alpha^i r_0^{(i)}$ $\beta = \sum_i \alpha^i \beta^{(i)}$

$z_0 = 1 + \sum_i \alpha^i z^{(i)}$

$r_0(\alpha) = \beta(\alpha, \epsilon) \frac{1}{z_0} \frac{dz_0}{d\alpha} \cdot z_0$

$\Rightarrow r_0 = \beta \frac{dz_0}{d\alpha} \Rightarrow (\alpha r_0^{(1)} + \alpha^2 r_0^{(2)} + \alpha^3 r_0^{(3)})$

$= (\alpha \beta^{(1)} + \alpha^2 \beta^{(2)} + \alpha^3 \beta^{(3)})(z^{(1)} + 2\alpha z^{(2)} + 3\alpha^2 z^{(3)})$

α^1: $r_0^{(1)} = \beta^{(1)} z^{(1)} = -2\epsilon z^{(1)}$

α^2: $r_0^{(2)} = 2\beta^{(1)} z^{(2)} + \beta^{(2)} z^{(1)} = -4\epsilon z^{(2)} + 2\epsilon (z^{(1)})^2$

$= -4\epsilon \left(\frac{z_1^{(1)}}{\epsilon} + \frac{z_2^{(2)}}{\epsilon^2} \right) + 2\epsilon \left(\frac{z_1^{(1)}}{\epsilon} \right)^2$

r_0 也和 β 一样不含 $\frac{1}{\epsilon}$ 以及高次. $\Rightarrow r_0^{(2)} = -4\epsilon z^{(2)}$

$\Rightarrow r_0(\alpha) = -2\epsilon\alpha \frac{dz}{d\alpha}$

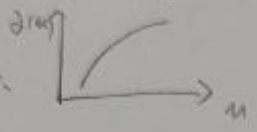
题3. $\alpha(u) = \dfrac{\alpha(Q)}{1+\alpha(Q)\frac{\beta_0}{4\pi}\log\frac{u^2}{Q^2}}$

QED:

取 $\alpha(Q)=\frac{1}{137}$，此时 $Q^2 \sim m_e^2$，$\beta_0 = -\frac{4}{3}$

则 $\alpha(u) = \dfrac{\alpha(Q)}{1-\frac{\alpha}{3\pi}\log\frac{u^2}{Am_e^2}}$ $\quad (A = e^{\frac{5}{3}})$

$u\uparrow$ $\alpha\uparrow$ $\therefore$



部分极点：when $1-\frac{\alpha}{3\pi}\log\frac{u^2}{Am_e^2}=0$ 时，$\alpha(u)$ 发散.

$\Rightarrow \frac{Q^2}{Am_e^2} = e^{\frac{3\pi}{\alpha}} \Rightarrow \frac{Q}{m_e} \sim 10^{280}$

对于SU(3)改成3

QCD: $\beta_0 = -\frac{4}{3}T_F n_q + \frac{11}{3}C_A$ (n_q = number of formions, $T_F=\frac{1}{2}$ $C_A=3$)

现实QCD中，$n_q \ll \frac{33}{2}$，$\therefore \beta_0 > 0$

则 $u\uparrow$ $\alpha\downarrow$

当u足够小，即 $1+\alpha(Q)\frac{\beta_0}{4\pi}\log\frac{\Lambda_{QCD}^2}{Q^2}=0$ 时，$\alpha(u)$ 发散

$\Rightarrow \Lambda_{QCD} \sim 1\,\text{GeV}$.

题4. $\Lambda_{QCD} = Qe^{-\frac{2\pi}{\alpha(Q)\beta_0}} \Rightarrow Q = \dfrac{\Lambda}{e^{-\frac{2\pi}{\alpha(Q)\beta_0}}}$

代入 $\alpha(u) = \dfrac{\alpha(Q)}{1+\alpha(Q)\frac{\beta_0}{4\pi}\log\frac{u^2 e^{-\frac{4\pi}{\alpha(Q)\beta_0}}}{\Lambda^2}} = \dfrac{\alpha(Q)}{1+\alpha\frac{\beta_0}{4\pi}\left(\log\frac{u^2}{\Lambda^2}-\frac{4\pi}{\alpha(Q)\beta_0}\right)}$

$= \frac{4\pi}{\beta_0}\dfrac{1}{\log(u^2/\Lambda^2)}$ 证毕.